

Memory Banking in 8086

Q. What is Memory Banking in the 8086 microprocessor? Explain the roles of the A0 and BHE# signals in accessing Even and Odd memory banks for byte and word transfers.

 **Definition to Remember**

Memory Banking is the hardware organization of the 8086's 1 MB physical memory into two parallel 512 KB halves: the **Even Bank** (connected to D0-D7) and the **Odd Bank** (connected to D8-D15). This allows the CPU to efficiently read/write either 8-bit bytes or 16-bit words.

1. Concept of Memory Banking

To support a 16-bit data bus while maintaining byte-level addressing, memory is split:

- * **Even Bank (Lower Bank):** Contains even addresses (00000H , 00002H). Connected to the lower half of the data bus (D0 - D7).
- * **Odd Bank (Upper Bank):** Contains odd addresses (00001H , 00003H). Connected to the upper half of the data bus (D8 - D15).

2. Role of A0 and BHE# Signals

The CPU uses two specific signals to select which bank to activate during a memory cycle:

- * **A0 (Address Line 0):** Selects the **Even Bank** (active when A0 = 0).
- * **BHE# (Bus High Enable):** Selects the **Odd Bank** (active low, BHE# = 0).

3. Byte and Word Access Scenarios

BHE#	A0	Selection	Transfer Type
0	0	Both Banks	Read/Write 16-bit Word from an Even Address. Uses D0-D15. Completes in 1 Clock Cycle .
0	1	Odd Bank	Read/Write 8-bit Byte from an Odd Address. Uses D8-D15.
1	0	Even Bank	Read/Write 8-bit Byte from an Even Address. Uses D0-D7.
1	1	None	No memory access.

4. Accessing a Word from an Odd Address (Unaligned Access)

If the CPU tries to read a 16-bit word starting at an **odd address** (e.g., 00001H), it cannot do it in one cycle. It requires **two clock cycles**:

- * **Cycle 1 (BHE#=0, A0=1):** Reads the lower byte from the odd address (00001H) via D8-D15.
- * **Cycle 2 (BHE#=1, A0=0):** Increments address to the next even boundary (00002H) and reads the upper byte via D0-D7.

★ **Must-Write Points (for 10 marks)**

1. **Memory Banking** splits 1 MB memory into two 512 KB banks: **Even** and **Odd**.
2. **Even Bank**: Even addresses, connected to lower data bus (**D0-D7**).
3. **Odd Bank**: Odd addresses, connected to upper data bus (**D8-D15**).
4. **A0=0** enables the Even Bank; **BHE#=0** enables the Odd Bank.
5. **BHE#=0, A0=0**: Reads a 16-bit word from an Even address in **1 clock cycle**.
6. Reading an 8-bit byte from Even sets A0=0, BHE#=1 (and vice versa for Odd).
7. Reading a 16-bit word from an **Odd Address** (unaligned) takes **2 clock cycles**.

⚡ **Quick Recall**

Memory Banking → 2 Banks (Even/Odd) → Even Bank (D0-D7, A0=0) → Odd Bank (D8-D15, BHE#=0) → Both=0 =
16-bit transfer in 1 cycle → Unaligned Word = 2 cycles